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# Rab29 fast exchange mutants: characterization of a challenging Rab GTPase

Rachel C. Gomez, Edmundo G. Vides and Suzanne R. Pfeffer*

Department of Biochemistry, Stanford University School of Medicine, Stanford, CA 94305-5307

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*Corresponding author: pfeffer@stanford.edu

**Summary/Abstract**

Rab29 has been implicated in multiple membrane trafficking processes with no described effectors or regulating proteins. Its fast nucleotide exchange rate and inability to bind GDI in cytosol make it a unique and poorly understood Rab. Because the conventional, “GTP-locked” Rab mutation does not have the desired effect in Rab29, we present here the use of a fluorescence-based assay to characterize novel mutants that display faster nucleotide exchange rates, allowing for GEF-independent Rab29 activation.


**Introduction**

Rab GTPases undergo a cycle between active, GTP bound forms and inactive, GDP-bound forms that interconvert by GTP hydrolysis and subsequent nucleotide exchange [1]. For most Rabs, the rate-limiting step in this cycle is GDP release [1]. Rabs have many chaperones and regulators to modulate their activity. For example, GDI (GDP dissociation inhibitor) binds preferentially to GDP-bound Rabs and solubilizes their prenyl groups [2]. GEFs (guanine nucleotide exchange factors) stabilize Rabs on membranes by activating them and driving subsequent effector binding and stabilization. GEF binding remodels the Rab nucleotide binding site, accelerating GDP dissociation, allowing for the more abundant GTP to bind [3]. GAPs (GTPase activating proteins) inactivate Rabs by stimulating their intrinsic GTP hydrolysis capacity [4]. However, to date, many Rabs have no reported GEFs or GAPs.

To study Rab phenotypes and binding partners, mutants are often used. A “GTP-locked” mutant, equivalent to Q67 in Rab1, drastically slows GTP hydrolysis [5]. This mutant aids assays of Rab function and binding partners that prefer the GTP bound Rab. Alternatively, mutations in the G1 motif, equivalent to S21N in Rab9, render Rabs GDP-preferring [cf. 6]. However, these mutations do not always function as expected [3].

Rab29 is localized on the trans Golgi network and activates Parkinson's Disease associated LRRK2 kinase [7]. It has also been implicated in ciliogenesis, retrograde endosomal membrane trafficking, Golgi structure, and T-cell receptor trafficking [8,9]. However, thus far, Rab29 has no reported GEF, GAP, or effector partners. Additionally, unlike all other Rab GTPases studied to date, Rab29 is not bound to GDI in cytosol and it releases GDP much faster than Rab5 [10]. Moreover, the Q67L "GTP-locked" Rab29 mutant does not behave as expected: unlike the wild type protein, it fails to localize to the Golgi and binds nucleotide only weakly [11]. To better understand Rab29's function and to discover its binding partners, we created "fast exchange" mutants that exchange nucleotide quickly and may be active in a GEF-independent manner. These mutants were designed based on a predicted structure of Rab29, with the goal of weakening interactions between the Rab and nucleotide.

We reported previously that Rab29 D63A binds nucleotide extremely poorly [10]. The D63 residue is conserved throughout the Ras superfamily and coordinates a water molecule adjacent to the $Mg^{2+}$ ion that holds the nucleotide in place. To characterize additional, similar mutants, we employed a fluorescence-based nucleotide release assay based on work from David Lambright and colleagues [12] and prior work from Roger Goody and colleagues [13] that we describe here. N-Methylanthranioyl (MANT) is a fluorescent modification that can be made to guanine nucleotides; its fluorescence increases when bound to a GTPase and use of MANT-GDP allows for continuous monitoring of nucleotide exchange upon addition of an excess of unlabeled nucleotide.

## 2. MATERIALS

### 2.1 Purifying Rab29 and its mutants

1. His-Sumo Rab29 WT and mutants: human Rab29 coding sequence (full length) with a 6X histidine-Sumo tag flush at the N-terminus in pET15b.
2. BL21 DE3 chemically competent expression cells.
3. Luria broth (LB): 10g/L bacto-tryptone, 5g/L yeast extract, 10g/L NaCl.
4. Carbenicillin (2000X): 100mg/mL in water, stored at -20°C.
5. 1M isopropyl beta-D-thiogalactopyranoside (IPTG); stored at -20°C.
6. PMSF: 100mM in 100% ethanol; stored at -20°C.
7. Protease inhibitors (100X): 100µg/ml each aprotinin, leupeptin, and pepstatin A; stored at -20°C.
8. Lysis buffer: 50mM Hepes pH 7.4, 200mM NaCl, 5mM $MgCl_2$, 5mM imidazole, 0.5mM DTT, 20µM GTP, 10% glycerol, PMSF, protease inhibitor mix.
9. Elution buffer: 50mM Hepes pH 7.4, 200mM NaCl, 5mM $MgCl_2$, 150mM imidazole, 0.5mM DTT, 20µM GTP, 10% glycerol.
10. Ni-NTA agarose.
11. 1ml syringe with frit at the bottom.
12. PD MiniTrap G-25.

### 2.2 MANT-GDP assay

1. Water bath.
2. 5X Exchange buffer: 100mM Hepes pH 7.4, 750mM NaCl, 25mM EDTA.
3. Glycerol.

4. Rab concentrated at or above 7.2μM.

5. 5mM MANT-GDP.

6. 2M $MgCl_2$.

7. Zeba 7K 0.5mL spin column.

8. 2X Desalting buffer: 60mM Hepes pH 7.4, 150mM NaCl, 5mM $MgCl_2$.

9. 1X Desalting buffer: 30mM Hepes pH 7.4, 150mM NaCl, 5mM $MgCl_2$.

10. 500mM EDTA pH 8.0.

11. 96-well black bottom plate.

12. Plate reader (Tecan 200 Pro).

13. 10mM GDP.

## METHODS

### 3.1 Purification of Rab29 proteins

1. A 10ml overnight culture of each Rab29 construct grown in LB supplemented with carbenicillin is used to inoculate a 1L culture of LB supplemented with antibiotics.

2. The culture is grown at 37°C until it reaches an $OD_{600}$ of ~0.6. The cultures are then transferred to 18°C and induced by adding IPTG to a final concentration of 0.3mM. Cultures are incubated at 18°C for an additional 16h.

3. Cells are collected by centrifugation (2700xg, 20min, 4°C). The pellet can be resuspended in PBS and repelleted. The washed pellet can be snap frozen in liquid nitrogen and stored at -20°C for later processing, if desired.

4. The pellet is resuspended in 30mL lysis buffer. The cells are broken by passing once through an EmulsiFlex-C5 apparatus at 20,000psi (Avestin).

5. The homogenate is transferred to chilled Oakridge centrifuge tubes and spun at 13,000xg for 20min at 4°C. Clarified supernatant is saved for analysis.

6. Clarified supernatant is added to fresh 50mL conical tubes and incubated with 30µL nickel-NTA resin (equilibrated in lysis buffer) for 2h at 4°C with rotation.

7. Flowthrough is collected for analysis. Resin is washed in the conical 3 times with a total of 60 column volumes lysis buffer.

8. Resin is transferred to an empty column (either 1ml syringe or larger column if necessary) using 500µL of lysis buffer.

9. Rab is eluted from resin with 500µL of elution buffer. Samples collected for analysis by SDS-PAGE.

10. Protein is desalted using a PD-10 G25 MiniTrap. Elution from desalting column is collected in 4 fractions. Fractions are assayed for protein concentration by Bradford assay and the most concentrated fractions are pooled and snap frozen in liquid nitrogen and stored at -80°C. See notes 1, 2.

**3.2 MANT-GDP assay**

1. Set up reaction mixture as follows: 1X Exchange buffer, 5µM Rab, 8% glycerol, 125µM MANT-GDP in a final volume of 100µL. Wrap tube in aluminum foil.
2. Incubate in a 25°C water bath for 2h.
3. Equilibrate Zeba mini trap desalting column in 1X desalting buffer while there is ~10min left in the nucleotide exchange reaction.
4. Quench reaction by placing on ice and adding a final concentration of 40mM $MgCl_2$ (2µL of 2M $MgCl_2$).

5. Desalt proteins using the equilibrated Zeba mini trap desalting column. After application of sample onto column, add the recommended 15µL stacker.
6. Set up the plate on ice with 100µL final sample volume. Set up reaction in duplicate and have duplicate reactions with EDTA for a control. Use 2X desalting buffer: For samples without EDTA: 30mM Hepes pH 7.4, 150mM NaCl, 5mM $MgCl_2$; For samples with EDTA: 30mM Hepes pH 7.4, 150mM NaCl, 5mM $MgCl_2$, 20mM EDTA.
7. Add 1µM final concentration of Rab and immediately begin monitoring in the plate reader. Excite samples at 360nm and monitor fluorescence at 440nm, scanning every minute and shaking before each measurement.
8. Wait until fluorescence has stabilized (~15-20 min). Continue as soon as measurement has stabilized to maximize protein activity and fluorescence.
9. Add 2µL of 20mM GDP for a final concentration of 200µM GDP. See note 4.
10. Place plate back in plate reader and continue measurements. It's important to work quickly here to observe the initial kinetic measurements.
11. Plot and analyze in Prism, considering the last reading before addition of the unlabeled GDP as time zero. Fit to a one phase decay.

**NOTES**

1. Yield from 1L culture for His-Sumo Rab29 is ~0.5mg/L; note that nucleotide is maintained throughout and all efforts are made to work quickly in the cold; proteins are snap frozen and not repeatedly frozen and thawed.

2. His-Sumo tag may be cleaved from Rab29 using His-SENP1 protease; the presence of the tag aids in soluble protein yield but is not essential; preps of untagged Rab29 yielded 1.2mg/8L culture.

3. If preferred, add 20uL of 1mM GDP using a multichannel pipet.

## FIGURE LEGENDS

Fig. 1. Purified Rab29. SDS-PAGE gel of the indicated proteins after purification. Impurity at ~20kDa may be cleaved His-Sumo.

Fig. 2. Rab mutants I64T and V156G show fast nucleotide dissociation. Fluorescence normalized to time point 0 for wild type (WT), I64T, and V156G Rab proteins.

Fig. 3. Rab29 mutants I64T and V156G show increased nucleotide release rates compared with wild type Rab29 (WT). Shown are calculated half-times from the fit of the MANT-GDP assay fluorescence data to a pseudo-first order decay.

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Fig 1

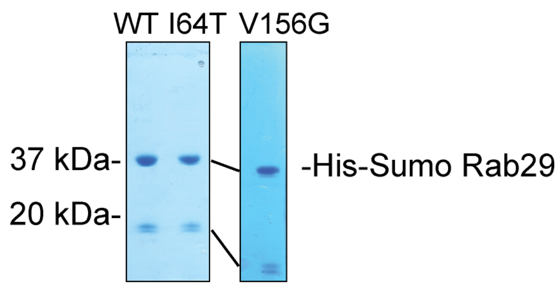


Fig 2

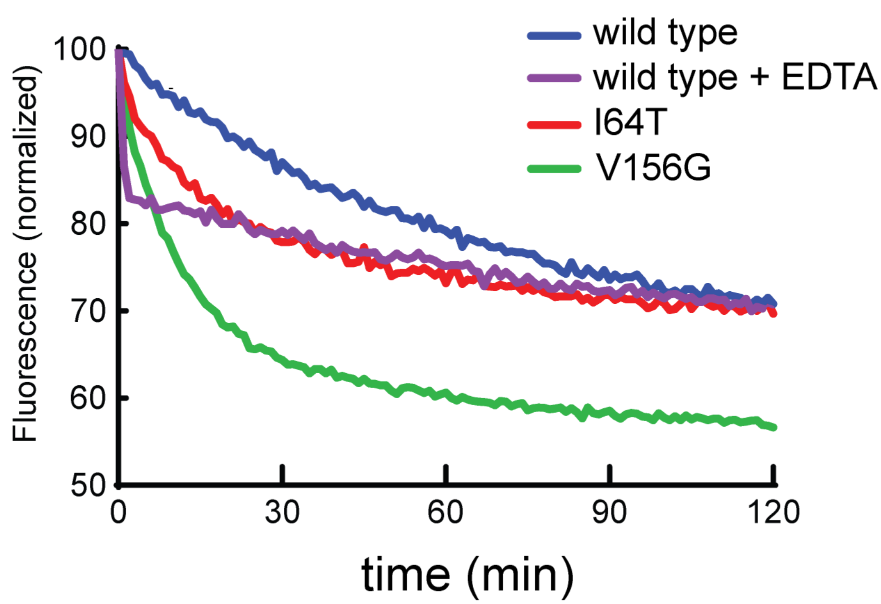


Fig 3